

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- D1 cannot work Night shift
- D2 must work before D3 (Morning < Afternoon < Night)
- Only one doctor per shift
- D3 cannot work Morning
- All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles
- Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I K.Abhishek dass/192212301 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1. Backtracking Search – Doctor Shift Assignment

Given

Doctors:

- D1
- D2
- D3

Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor gets exactly one shift.

Step 1: Start Backtracking

Initially:

Doctor Shift

D1 ?
D2 ?
D3 ?

Available shifts = {Morning, Afternoon, Night}

Step 2: Assign D1

Try Morning.

Assignment:

D1 → Morning ✓

Remaining shifts:

{Afternoon, Night}

No constraints violated.

Step 3: Assign D2

Try Afternoon.

Assignment:

D2 → Afternoon ✓

Remaining shift:

{Night}

Check constraints:

- D2 before D3 ✓ (possible)

Continue.

Step 4: Assign D3

Remaining shift = Night

Assignment:

D3 → Night

Check constraints:

- D3 not Morning ✓
- D2 before D3
- Afternoon before Night ✓

- One doctor per shift ✓

All constraints satisfied.

Search Tree

```
Start
|
|-- D1 = Morning
      |
      |-- D2 = Afternoon
            |
            |-- D3 = Night
                  ✓ Goal Found
```

Constraint Checking

Step	Constraint	Status
D1 = Morning	D1 not Night	✓
D2 = Afternoon	Before D3 possible	✓
D3 = Night	D3 not Morning	✓
Final	One doctor per shift	✓
Final	Every doctor assigned	✓

Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

2. BFS Search (5 × 5 Grid)

Assume the following grid:

```
S  .  .  X  .
.  X  .  X  .
.  X  .  .  .
.  .  X  X  .
.  .  .  .  G
```

Legend

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Coordinates

S = (0,0)

G = (4,4)

BFS Algorithm

Uses a **Queue (FIFO)**.

Initial Queue

Queue = [(0,0)]

Visited = {(0,0)}

Step 1

Expand

(0,0)

Possible moves

- Right → (0,1)
- Down → (1,0)

Queue

[(0,1), (1,0)]

Step 2

Expand

(0,1)

New node

(0,2)

Queue

[(1,0), (0,2)]

Step 3

Expand

$(1, 0)$

New node

$(2, 0)$

Queue

$[(0, 2), (2, 0)]$

Step 4

Expand

$(0, 2)$

New node

$(1, 2)$

Queue

$[(2, 0), (1, 2)]$

Step 5

Expand

$(2, 0)$

New node

$(3, 0)$

Queue

$[(1, 2), (3, 0)]$

Continue similarly until Goal.

Optimal Path Found

$(0, 0)$

↓
(1, 0)
↓
(2, 0)
↓
(3, 0)
→
(3, 1)
↓
(4, 1)
→
(4, 2)
→
(4, 3)
→
(4, 4)

Manhattan Distance Heuristic

Formula

$$\begin{bmatrix} h(n) = |x_g - x| + |y_g - y| \end{bmatrix}$$

Example

At

(2, 3)

Goal

(4, 4)

$$\begin{bmatrix} h = |4 - 2| + |4 - 3| \end{bmatrix}$$

$$\begin{bmatrix} = 2 + 1 = 3 \end{bmatrix}$$

Total Cost

Each move costs **1**.

Number of moves = **8**

Therefore

Optimal Cost = 8

Final Answer

Optimal Path

S
↓
↓
↓
→
↓
→
→
→
G

Total Cost

8

3. Autonomous Rescue Robot

(a) Constraint Analysis

1. Movement (Up, Down, Left, Right)

The robot is restricted to four directions. Diagonal movement is not allowed, increasing planning complexity.

2. Obstacles

Blocked rooms cannot be entered.

The robot must search for alternative paths.

3. Limited Sensing

The robot only observes adjacent rooms.

It cannot know the entire environment initially.

The map is updated as it explores.

4. Movement Cost

Normal movement

Cost = 1

Risky zone

Cost = 1 + 2 = 3

The robot prefers safer routes even if they are slightly longer.

5. Dynamic Knowledge

As the robot discovers blocked paths, it replans its route.

This makes it an **online search problem**.

(b) Solution Approach

Appropriate Strategy

Uniform Cost Search (UCS) combined with an Online Search Agent

Steps

Step 1

Start at S.

Step 2

Observe adjacent cells.

Step 3

Add reachable cells to the priority queue based on cumulative cost.

Step 4

Choose the lowest-cost node.

Step 5

If a new obstacle is detected,

- Update the map.
- Recalculate the best path.

Step 6

Repeat until Goal is reached.

Example

Start

↓

Observe neighbours

↓

Choose cheapest path

↓

Move

↓

Update knowledge

↓

Repeat

↓

Goal

(c) AI Concepts

Intelligent Agent

The rescue robot acts as an intelligent agent.

It

- senses the environment,
 - reasons about available paths,
 - takes actions,
 - learns new information.
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Rationality

A rational agent always chooses the action with the minimum expected total cost.

Example:

Instead of entering a risky room (cost = 3),

it may take a longer safe route (cost = 1 per move) if the total cost is lower.

Environment Type

The environment is:

Property	Description
Partially Observable	Robot sees only nearby cells
Dynamic	New obstacles may be discovered
Sequential	Current actions affect future states
Discrete	Grid consists of separate rooms
Single Agent	Only one robot is making decisions

(d) Justification

Uniform Cost Search with an Online Search Agent is the most suitable approach because:

1. It guarantees the minimum-cost path.
 2. It handles different movement costs (normal and risky zones).
 3. It dynamically updates the path when new obstacles are discovered.
 4. It works effectively in partially observable environments.
 5. It enables safe and efficient rescue operations.
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